

A Distributional Perspective on Forward Guidance and Monetary Policy along the US Term Structure*

Péter Horváth

Abstract

This paper investigates the impact of monetary policy and forward guidance on the U.S. term structure of interest rates with a focus on how it affects conditional volatility dynamics. Employing a dynamic location-scale model and several high-frequency surprises, I find that monetary policy announcements have a significant effect on the conditional volatility of the yield curve. While standard policy rate shocks generally increase volatility, forward guidance tends to compress it. However, these effects on conditional volatility are far smaller and less persistent than the conditional mean effects. The volatility channel is primarily driven by information signals and is asymmetric with respect to the sign of the shock. The results are robust to alternative specifications.

Keywords: Forward guidance, Term Structure, Conditional volatility

JEL: E43, E52, C32, G14

*I am grateful to Gregor von Schweinitz and Michele Lenza for their excellent comments on the paper.

1 Introduction

Since the Global Financial Crisis, central banks have increasingly relied on communication as an active policy instrument. In the United States, forward guidance became a central tool once the policy rate approached its effective lower bound (ZLB). Because the current policy rate could no longer be lowered, communication about the future course of policy action provided further easing to the stance of monetary policy. The Federal Reserve’s forward guidance evolved over time from relatively simple, less informative statements to state-contingent language that explicitly tied future policy action to macroeconomic variables like the unemployment rate. As forward guidance proved useful in managing expectations, it remained a permanent fixture in the monetary toolkit, helping to smooth the gradual rate increases that began in December 2015 and continuing to shape market expectations today.

Following the definitions of Campbell et al. (2012), central banks manage expectations through two distinct mechanisms. First, the central bank can make commitments regarding future policy action, referred to as Odyssean guidance. Second, it can communicate its own forecasts about the economic outlook, which inherently implies a future course of monetary policy, known as Delphic guidance. While the intentions behind these interventions are clear, their transmission mechanisms are theoretically complex. For example, Hanson and Stein (2015) show that guidance indicating persistently low interest rates can paradoxically raise long-term yields if markets interpret it as a signal of stronger future growth and inflation. Similarly, Andrade et al. (2019) argue that commitments to maintain low rates may be interpreted either as bad news about future economic conditions or as good news reflecting a more accommodative policy stance. Consequently, the effects of both Odyssean and Delphic guidance on broader economic uncertainty remain theoretically ambiguous—positive and negative signals may either reduce or amplify market volatility.

To study how different monetary policy tools affect financial markets and the economy, recent literature relies heavily on high-frequency identification (HFI) methods. Asset price movements in tight windows around FOMC meetings capture exogenous policy surprises. First established by Kuttner (2001) and Bernanke and Kuttner (2005), this approach was expanded by Gurkaynak et al. (2005) (GSS), who demonstrated that at least two factors are needed to explain the co-movement of yields at different maturities. A “Target” factor that captures current policy rate shocks from high-frequency changes in shorter term contracts and a “Path” factor that captures future policy (forward guidance) through high-frequency changes in medium-term contracts. Swanson (2021) extended this framework to show that forward guidance and large-scale asset purchases (LSAPs) operate primarily through a signaling channel, lowering the entire term structure of expected future short-term rates. The most active frontier of this HFI literature concerns central bank information effects (Nakamura and Steinsson (2018); Jarociński and Karadi (2020)). If a contractionary shock conveys superior central bank information about stronger future economic conditions, it can generate seemingly puzzling responses in macro-financial variables. Recent work has focused heavily on disentangling these “pure” policy shocks from central bank information and the Fed’s response to incoming news (Bauer and Swanson (2023); Jarocinski and Karadi (2025)).

However, by primarily focusing on the conditional mean responses of yields and asset prices, the HFI literature risks overlooking a crucial second-order dynamic. Policy communication may shape expectations through two distinct channels: it can depress the expected path of future interest rates (moving the conditional mean), but it can also provide greater certainty to markets by clarifying the future course of policy (affecting higher-order moments, specifically conditional volatility). Policy communication is frequently described in practice as a way to reduce uncertainty and avoid “surprising” markets. Filardo and Hofmann (2014) explicitly note that forward guidance can reduce uncertainty and interest rate volatility, potentially lowering risk premia.

Addressing this conditional volatility channel is vital. The macro-uncertainty literature already demonstrates that uncertainty is a significant driver of economic downturns (see e.g.: Bloom (2009); Jurado et al. (2015); Baker et al. (2016)). More specifically, Husted et al. (2020) show that uncertainty about monetary policy itself causes economic contractions, implying that alleviating this uncertainty might offset the contractionary effects of rate hikes. Bundick et al. (2017) and Bundick et al. (2024) provide direct evidence that forward guidance operates partly by changing the term structure of policy rate uncertainty, which helps explain yield movements beyond standard monetary policy shocks. If monetary policy can alleviate such uncertainty, this constitutes an important transmission mechanism in its own right. Conversely, if policy actions increase uncertainty, policymakers face a mean-variance trade-off where intended policy effects are partly offset by adverse volatility.

This paper studies forward guidance from a distributional perspective, asking whether forward guidance affects not only the location of the distribution of U.S. Treasury yields, but also its scale (i.e., the conditional volatility that investors face when pricing and hedging interest-rate risk). Even if average yield effects appear modest at certain maturities, systematic reductions in conditional volatility can materially affect term premia, portfolio allocation, and broader financial conditions.

This distributional view connects naturally to the growing “outlook-at-risk” literature in macro-finance (see e.g.: Adrian et al. (2019); Lopez-Salido and Loria (2024); Furceri et al. (2025)). This literature emphasizes that some factors (such as financial stress) impact macro aggregates asymmetrically across their distributions. Specifically, financial stress affects the left tail of GDP growth and the right tails of inflation and debt more adversely than it does the medians. However, Frangiamore et al. (2025) shows that fiscal consolidation also affects the tails asymmetrically. The paper derives the tail effects from how fiscal policy shocks affect the conditional location and the conditional scale of the debt distribution. Their dynamic model shows that a contractionary fiscal policy shock can reduce both the conditional location of future public debt and its conditional scale too (in other words reducing debt-uncertainty).

To provide this distributional view on monetary policy and forward guidance, I closely follow the methodologies of two papers, Hubert and Labondance (2018) and Frangiamore et al. (2025). I extend the framework of Hubert and Labondance (2018) in three key dimensions. First, rather than modeling conditional heteroskedasticity purely as an ARCH-type process, I estimate a location–scale specification in which forward guidance, monetary policy shocks, and macro-financial controls can independently shift both the conditional mean and the conditional volatility of yields across the term structure. Second, I embed this

model in a local projections framework similar to the one used by Frangiamore et al. (2025) in order to trace the dynamic responses (both in terms of location and scale) of different maturity yields. Third, I use novel methods from the HFI literature to capture different components on monetary policy, namely: standard policy rate shocks, Odyssean forward guidance shocks and Delphic forward guidance shocks.

2 Data

The empirical analysis primarily uses daily zero-coupon U.S. Treasury yields across multiple maturities from the Gürkaynak et al. (2007) dataset. In order to capture the full shape of the yield curve, I use maturities ranging from 6 months to 10 years. This allows for investigating how different monetary policy shocks influence short-term instruments as well as longer term expectations about policy stance and the economy. The Affine Term Structure Model of Christensen et al. (2011) decomposes the nominal yields into Expected Short Rates (ESR) and Term Premia (TP), which allows for individual examination of how the conditional volatility of risks and expectations influence the overall response of nominal yields. Lastly, I retrieve daily instantaneous forward rates on Treasury Inflation Protected Securities (TIPS) and Breakeven Inflation rates to check if the volatility channel has any spillover to real economy expectations. Summary statistics of each yield curve are reported in Table 1 in the Appendix.¹

In addition to the yield curve data, the specification includes a set of macro-financial control variables capturing global risk factors, the monetary policy stance, financial market stress and uncertainty, as well as proxies for real economic activity. A more detailed description of these variables, along with summary statistics, is provided in Table 3 in the Appendix. Since shocks in the HFI literature are constructed to be plausibly exogenous, the inclusion of these controls is not essential for the consistent estimation of the location effects. However, improving the fit of the location model is important for the precision of the scale estimates, as conditional volatility is proxied by the absolute residuals from the mean model.

2.1 Monetary policy and forward guidance shocks

Monetary policy shocks are constructed by replicating alternative identification schemes using the U.S. Monetary Policy Data (USMPD) database introduced by Acosta et al. (2025).² All HFI shocks are computed using data from 30-minute event windows around FOMC statement releases. While the USMPD also allows for the construction of shocks for windows around press conferences and meeting minutes, these alternative event windows are not used in the baseline analysis and are left for future research.

As a baseline, I adopt the identification strategy of Jarociński (2024), which yields four distinct shocks: a target rate shock, an Odyssean forward guidance shock, a Delphic information shock, and a large-scale asset purchase (LSAP) shock. In this paper, I focus on the first three factors. To assess robustness, I also

¹Data on Nominal Yields, Expected Short Rates and Term Premia are retrieved from the San Francisco Fed website and available at <https://www.frbsf.org/research-and-insights/data-and-indicators/treasury-yield-premiums/>. Data on real yields and breakeven rates are retrieved from the Fed Board through <https://db.nomics.world/>

²Retrieved from the San Francisco Fed website and available at <https://www.frbsf.org/research-and-insights/data-and-indicators/us-monetary-policy-event-study-database/>.

consider alternative HFI shock measures from Rogers et al. (2018) and Jarociński and Karadi (2020). The “path” factor from the former and the central bank information shock from the latter closely correspond to the Odyssean and Delphic factors respectively. All shock series are scaled to have unit standard deviation. The sample period runs from January 2004 to May 2026. Descriptives and further details for the shocks are available in Table 2 of the Appendix.

3 Methodology

The empirical framework builds on the location–scale model of Rigby and Stasinopoulos (2005), in which the conditional distribution of a dependent variable is characterized by separate equations for its location and scale parameters. In its general form,

$$y \mid X \sim \mathcal{D}(\mu, \sigma),$$

$$\mu = \beta_0 + X\beta,$$

$$\log \sigma = \gamma_0 + X\gamma.$$

where μ governs the conditional mean and σ the conditional dispersion of the dependent. Applied to yield changes, this framework allows forward guidance and monetary policy shocks to influence not only the expected response of interest rates, but also the degree of uncertainty surrounding that response. Within this framework, the coefficient vector β captures the effects of a regressor on the level (“location”) of the dependent variable and is directly comparable to coefficients estimated via OLS in standard event-study regressions. The coefficient vector γ captures how a regressor shifts the conditional volatility (“scale”) of the dependent variable. A negative value implies a compression of the conditional distribution, i.e. a reduction in uncertainty, while a positive value indicates an expansion of the conditional distribution and higher uncertainty.

To trace out the dynamic responses, I estimate location-scale models using the local projections framework of Jordà (2005). This specification allows me to trace both the response of yields and the response of their conditional volatility at different forecast horizons h . At each horizon, the regressions are estimated using a two-step OLS approach, where conditional volatility is proxied by the absolute residuals from the location equation. This setup closely follows Frangiamore et al. (2025), who estimate the scale effects of fiscal shocks on public debt.

Daily financial market data are considerably noisier than annual macro aggregates, and volatility clustering is therefore a concern. Engle (1982) and Bollerslev (1986) provide the canonical framework for modeling time-varying conditional variance. Hubert and Labondance (2018) make a similar point and uses ARCH-augmented dynamic event-study regressions to study how ECB forward guidance affects the euro yield curve. Building on that approach, I therefore augment the scale equation with an additional

ARCH term to capture volatility clustering³ in yields, and estimate the following system of regressions:

$$\begin{aligned} r_{t+h}^m - r_{t-1}^m &= \beta_0^h + \beta_1^h x_t + \beta_2^h z_t + \varepsilon_{t+h}^h, \\ |\varepsilon_{t+h}^h| &= \gamma_0^h + \gamma_1^h x_t + \gamma_2^h z_t + \gamma_3^h |\varepsilon_{t+h-1}^h| + u_{t+h}. \end{aligned}$$

In these regressions, x denotes the matrix of monetary policy and forward guidance shocks, z represents the set of control variables, and r^m denotes yields at maturity m . The models are estimated at horizons $h \in \{0, 1, 3, 5, 10, 15, 20, 25\}$, where $h = 0$ corresponds to the standard event-study specification and captures the response of yields and their conditional volatility on the announcement day, while $h = 25$ measures the response approximately one business month after the shock. All regressions are estimated using Newey–West heteroskedasticity and autocorrelation consistent standard errors. Results are reported with one standard deviation confidence bands.

Since the scale equation uses the absolute residuals from the location equation as a proxy for conditional volatility, both sets of impulse responses are expressed in the same units. Accordingly, the location coefficient, β^h , can be interpreted as: “A one standard deviation contractionary shock increases the yield by β^h percentage points at horizon h .” Similarly, the scale coefficient, γ^h , can be interpreted as: “A one standard deviation contractionary shock changes the conditional volatility of the yield by γ^h percentage points at horizon h .”

4 Results

This section presents the empirical results for the location and scale effects of monetary policy shocks on the term structure of interest rates in the US. I document the responses with respect to three shocks. A Target rate surprise which corresponds to a standard interest rate shock. An Odyssean forward guidance surprise which corresponds to unexpected changes in the expected future path of monetary policy arising from central bank commitment. Finally, a Delphic forward guidance surprise which corresponds to an information shock revealing private information of the Fed regarding economic outlook and its implications for future policy. The results are organized as follows. I first document the location and scale responses to contractionary realizations of each shock. I then examine whether the scale responses differ between contractionary and expansionary shocks. In order to better understand the scale effects of monetary policy on nominal rates, I run the location-scale regressions on Expected Short Rates and Term Premia separately. Lastly, I check for possible real expectation spillovers by running the regressions on instantaneous forward rates of TIPS yields and Breakeven Inflation.

Figure 1 reports the estimated location and scale impulse responses of Treasury yields to the three shocks.

³Introducing an ARCH term into the scale equation prevents the direct application of the Machado and Silva (2019) framework, which requires identical covariates in the location and scale equations to recover quantile-specific marginal effects. This does limit the investigation to some extent; however, this is of little consequence, as the focus of the analysis is on volatility responses themselves rather than on heterogeneous effects across quantiles of the yield distribution.

The location effects are consistent with the existing literature. Target rate shocks have the strongest effects at the short end of the yield curve; Odyssean forward guidance shocks raise yields across the entire term structure; Delphic information shocks increase short-term yields but lower yields at longer maturities. As these responses closely replicate the results of Jarociński (2024), they serve two primary purposes. First, they validate both the replicated shock series and the local projection framework. Second and more importantly, they provide a natural benchmark against which to assess the magnitude of the estimated scale responses.

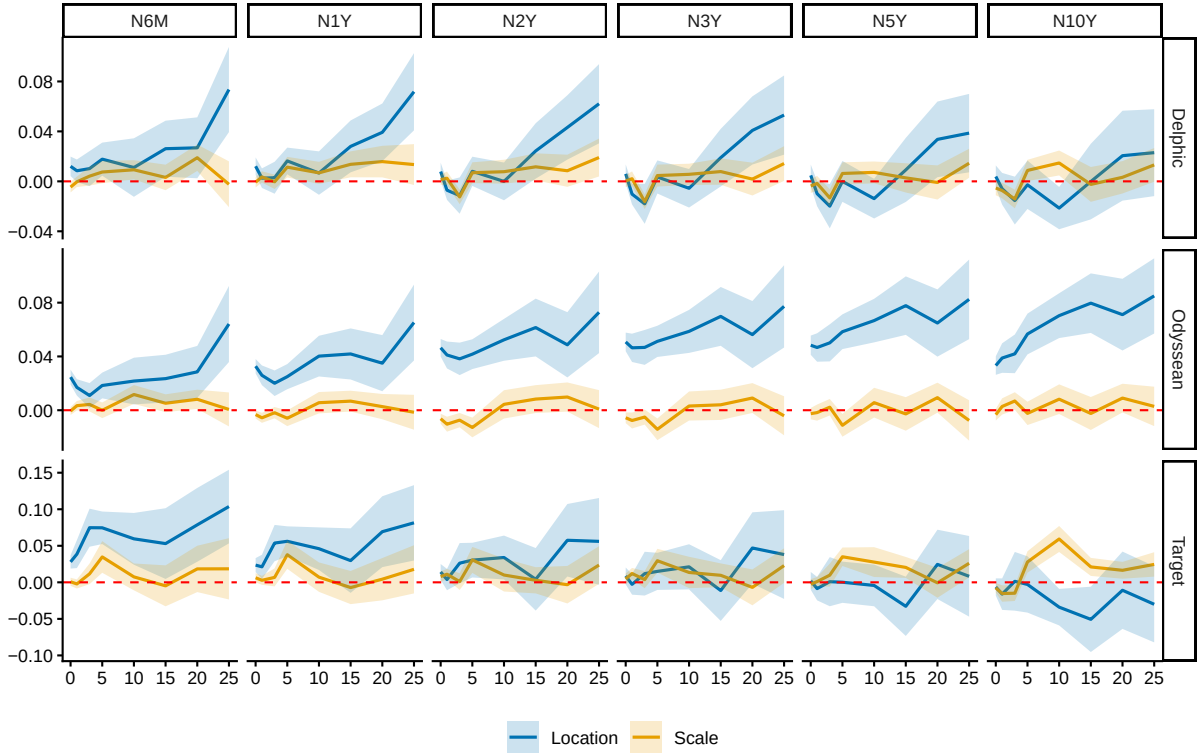


Figure 1: Dynamic responses of the nominal term structure to monetary policy shocks.

Relative to the location effects, the scale responses are considerably smaller in magnitude and substantially more short lived. The largest responses occur within the first few trading days following the announcement and dissipate rapidly thereafter. Subsequently, monetary policy announcements might influence uncertainty in Treasury markets only in the short term. However, despite the modest size and lack of persistence, the scale responses are systematically different across the three types of shocks. Target rate surprises generally increase conditional yield volatility across the term structure, with the largest response observed for the 10-year Treasury yield. In contrast, both forward guidance shocks tend to compress conditional volatility. The effects of Odyssean shocks are concentrated at medium maturities and are negligible at the short and long ends of the yield curve. Delphic information shocks also have negligible effects on shorter maturities, but impact both medium and long term yields.

This suggests two forces at play in the drivers of yield volatility around FOMC meetings: policy intervention causing uncertainty and communication mitigating it. Unexpected changes in the current policy rate increase uncertainty regarding both the prevailing and expected future stance of monetary policy, leading to higher conditional volatility across the entire term structure. On the other hand, communication shocks (whether reflecting commitment about rates or information about economic outlook) tend to reduce uncertainty at medium maturities. Consequently, forward guidance offsets the uncertainty caused by shocks to the policy rate around the middle of the yield curve. While communication effects are negligible at the short end, the volatility response of 6-month and 1-year yields to a policy rate shock is small in magnitude and fades within a few trading days.

Unlike short to medium maturities, the scale response of the 10-year yield to a policy rate shock is relatively large, persistent and is not offset by either Odyssean or Delphic communication. One possible explanation is that policy rate interventions convey two opposing signals for longer-term interest rates. On the one hand, market participants may anticipate policy normalization some time after the Fed raises rates. This could be supported by the small (but mostly insignificant) negative location response as well. On the other hand, an unexpected policy tightening may increase uncertainty surrounding the longer-run macroeconomic outlook, which prompts market participants to reassess their outlook as well. Since these macro fundamentals are more influential for longer-term yields, communication about medium-term future policy may be insufficient to offset the increase in uncertainty over longer horizons.

The analysis thus far has imposed symmetry between contractionary and expansionary shocks, implying that both types of announcements affect conditional volatility to the same extent, differing only in sign. While this is a common simplification in most empirical models, there is substantial evidence on the asymmetric effects of opposing sign shocks (see e.g.: Cover (1992), Tenreiro and Thwaites (2016), Angrist et al. (2018)). While the simplification of symmetry is reasonable for the location estimates, it is particularly restrictive for the estimation of scale responses. For example, both good and bad macroeconomic news should help coordinate market expectations, whereas the symmetric assumption mathematically constrains one to increase and the other to decrease the conditional volatility. The local projections, however, can be easily adapted to accommodate the potentially asymmetric effect of opposing sign shocks. To investigate, the equations are re-estimated after decomposing each shock into separate contractionary and expansionary components.⁴ The results in Figure 2 indicate that accounting for the sign of the shock matters for both the magnitude and the persistence in the scale responses.

⁴Expansionary shocks are multiplied by -1 so that both shock series have a consistent interpretation when plotting the impulse responses. The sign-dependent shocks are not re-scaled using the standard deviations of the respective subsets.

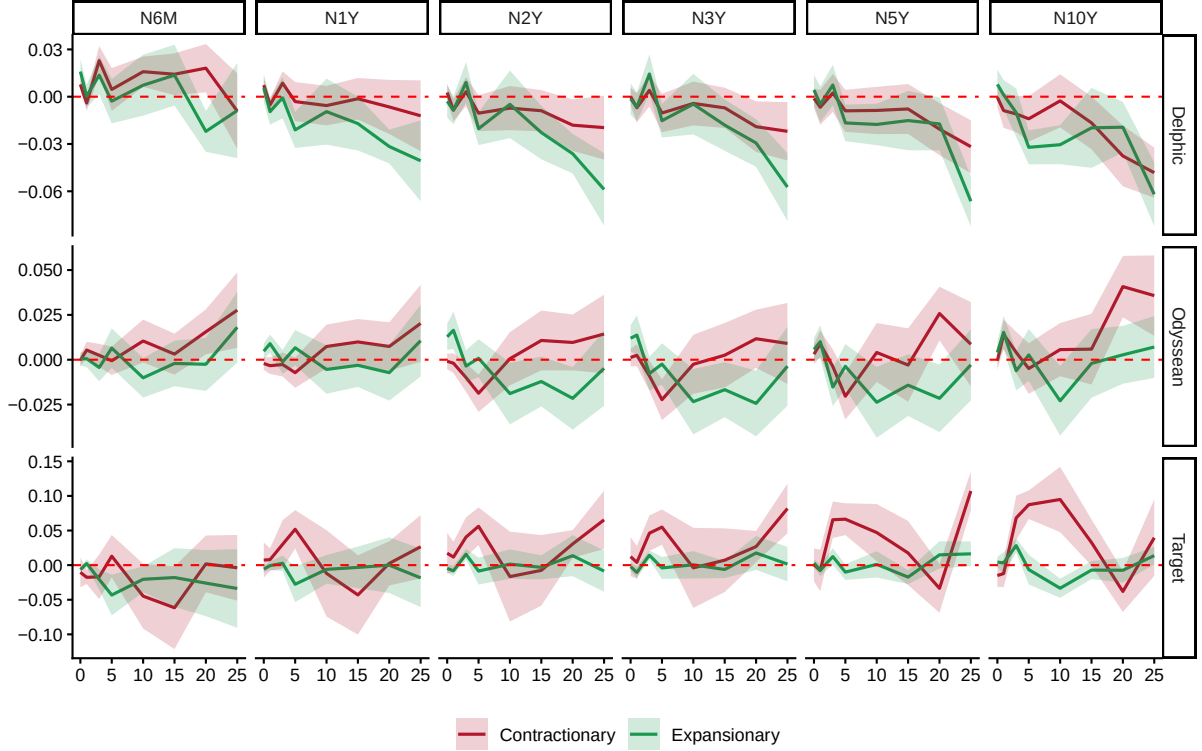


Figure 2: Sign-dependent scale effects across the nominal term structure of interest rates.

Asymmetry in the responses is present across all three shocks, though it manifests differently in each case. For Target shocks, contractionary surprises mirror symmetric estimates while expansionary ones barely affect volatility. Delphic shocks no longer yield an immediate volatility reduction. Instead, they generate a persistent compression at the long end, with positive signals slightly more effective than negative ones. Odyssean shocks exhibit a distinct temporal asymmetry. Contractionary forward guidance initially compresses volatility before the effect reverses, while accommodative commitments first increase, then lower volatility. These short-run effects are short-lived, however. At medium maturities, accommodative guidance suppresses volatility for about a month, yet at the long end, contractionary commitments raise volatility over the same horizon.

These results confirm that the volatility channel is primarily driven by information signals. This information can either be communicated directly, as in the case of Delphic forward guidance, or inferred by market participants from policy actions and commitments, as with Target and Odyssean shocks. Direct communication by the Fed regarding the economic outlook appears to be the most effective at reducing uncertainty irrespective of whether the news is good or bad. Odyssean guidance effectively coordinates medium-term expectations but fails to anchor the long run. Moreover, it may actually heighten long-term uncertainty, which reveals another policy trade-off.

4.1 Sources of the volatility response: Expected short rates versus term premia

The preceding analysis establishes that monetary policy announcements affect not only the expected level of Treasury yields but also their conditional volatility. A natural next question is which component of the yield curve drives these second-moment responses. Since nominal yields reflect both expectations about the future path of short-term interest rates and compensation for bearing interest rate risk, decomposing the yield curve provides additional insight into the economic channels through which monetary policy influences uncertainty. To this end, I run the location-scale regressions separately for expected short rates (ESR) and term premia (TP). The estimated scale impulse responses can be seen in Figure 3.

The nominal yields are decomposed into ESR and TP as in Christensen et al. (2011) such that $r^m = ESR^m + TP^m$. While this identity holds exactly for yield levels (and level responses), it does not generally extend to the estimated scale responses for two reasons. First, conditional volatility is proxied using the absolute residuals from the location equation, making the scale specification inherently nonlinear. Second, from a variance decomposition perspective $Var(r^m) = Var(ESR^m) + Var(TP^m) + 2 \cdot Cov(ESR^m, TP^m)$, where the covariance term is not separately identified in the present framework. Accordingly, the decomposition is intended to provide qualitative evidence on the economic sources of the volatility responses rather than an exact accounting identity.

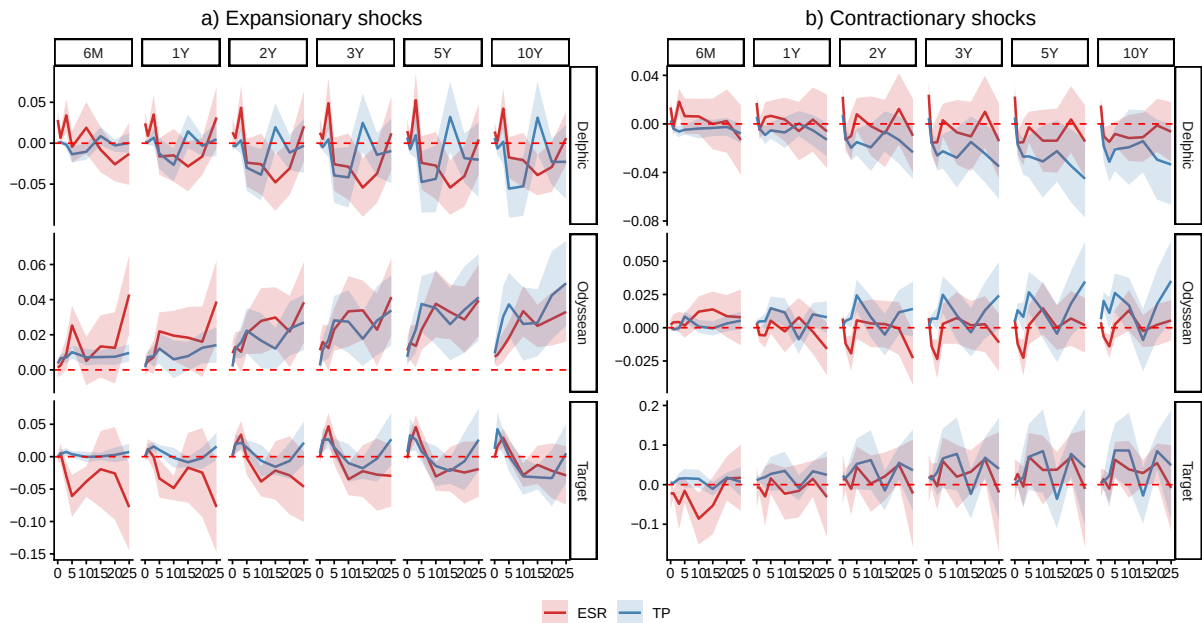


Figure 3: Scale responses of Expected Short Rates and Term Premia.

The transmission of Delphic shocks is clearest in the ESR-TP decomposition, as the scale responses of each component behave similarly to those of the nominal yields. This suggests that Federal Reserve com-

munication about economic outlook simultaneously reduces uncertainty surrounding the future path of monetary policy and the compensation investors require for bearing interest rate risk. Since the responses of the individual components co-move with the total, there is little indication that Delphic shocks alter the covariance structure between the two components and the volatility decline is primarily driven by the uncertainty reduction of each component. Economically, this means that Fed information reduces uncertainty without altering the way markets think about how future policy expectations and longer-term risks are linked.

The decomposition of scale responses to Target shocks is relatively straightforward as well. Contractionary policy surprises generally increase the conditional volatility of both expected short rates and term premia, with the only notable exception being the 6-month expected short rate, whose scale response is slightly negative. This broadly matches the pattern of responses observed in nominal yields. However, the confidence bands around the component responses are noticeably wider than those for nominal yields, despite the latter displaying a relatively clear increase in conditional volatility. This suggests that part of the joint volatility arises from stronger co-movement between ESR and TP following a contractionary shock. A possible explanation for stronger co-movement along with rising uncertainty in both components is that unexpected policy tightening could lead to revisions in inflation expectations accompanied by higher inflation uncertainty. Expansionary policy rate shocks, by contrast, have little discernible effect on either expected short rate or term premium volatility. The corresponding responses of nominal yields are likewise close to zero, indicating only limited evidence of a volatility channel following unexpected policy easing.

The most puzzling component-wise responses come from Odyssean shocks. Expansionary commitments to maintain lower future policy rates increase the conditional volatility of both ESR and TP over approximately the first month following the announcement. During the first few trading days, these responses are consistent with the temporary increase in nominal yield volatility documented earlier. At longer horizons, however, the conditional volatility of nominal yields declines despite both underlying components exhibiting higher volatility. This pattern is consistent with a substantial decline in the covariance between ESR and TP. A plausible explanation is the inherently state-contingent nature of forward guidance. While such guidance may increase the sensitivity of both interest rate expectations and investors' assessment of long-term risks to incoming macroeconomic information, it may also strengthen the negative relationship between the two components. For example, favorable news may lead markets to expect a steeper policy path (raising ESR) while simultaneously reducing the compensation required for bearing duration risk (lowering TP), with the opposite occurring following adverse news. As a result, ESR and TP might move in opposite directions more strongly (decline in covariance) and consequently reduce the conditional volatility of nominal yields. For contractionary Odyssean shocks, the short-run decline in ESR volatility and the subsequent increase in TP volatility broadly match the patterns observed for nominal yields.

4.2 Spillover to real yields and breakeven inflation

Thus far, the analysis has focused exclusively on nominal Treasury yields, which jointly reflect expectations about future real interest rates and inflation compensation. While this establishes how monetary policy announcements affect the overall term structure, it does not reveal whether the documented volatility responses originate from changes in uncertainty surrounding real financing conditions or inflation expectations. To address this question, I repeat the location-scale regressions using instantaneous forward TIPS yields and breakeven inflation rates. Figure 4 documents these scale impulse responses to expansionary and contractionary shocks.

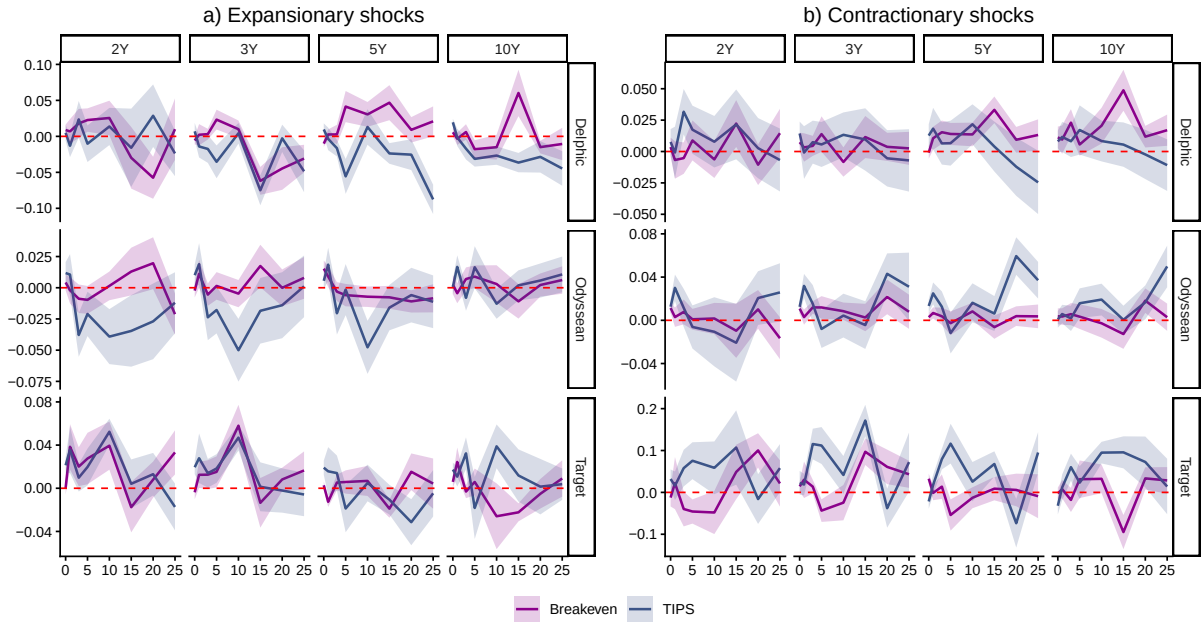


Figure 4: Scale responses of forward TIPS yields and breakeven inflation rates.

The scale impulse responses reveal that the volatility channel differs between real interest rates and inflation expectations. Policy rate shocks predominantly alter the conditional volatility of real rates, while the responses of expected inflation volatility are weaker and more heterogeneous. Both contractionary and expansionary Target shocks raise uncertainty surrounding future real interest rates, although the response is considerably stronger following policy tightening. Expansionary shocks primarily increase short-horizon inflation uncertainty, whereas contractionary shocks initially compress it before generating a modest increase at longer horizons.

Odyssean shocks have virtually no effect on inflation uncertainty. Instead, the response of real rates closely mirrors the patterns previously observed for nominal yields: commitments to a more accommodative future policy stance reduce uncertainty surrounding real financing conditions over the medium term, whereas commitments to tighter future policy increase it. By contrast, Delphic forward guidance has much more limited effects on the conditional volatility of TIPS yields and primarily affects the volatility

of inflation expectations. Overall, these findings suggest that commitment-based communication operates mainly by increasing certainty regarding future real financing conditions, whereas information about the economic outlook primarily induces revisions in inflation expectations.

4.3 An illustrative counterfactual exercise

The location and scale impulse responses documented above quantify how monetary policy and forward guidance shocks move the conditional mean and volatility of yields, but they do not by themselves indicate how much of the observed path of any given yield is attributable to these shocks. This section constructs a simple counterfactual to address that question directly, isolating how much of the 2-year nominal yield's evolution over the sample is attributable to the location and scale effects of Odyssean forward guidance shocks specifically.

Unlike Vector-Autoregressions, local projections are inherently not forecasting models, where simulations are readily available by the system of equations. It is only at $h = 0$ where a genuine one-step-ahead forecast of $r_t - r_{t-1}$ is computed, whereas the remaining horizons $h > 0$ are separate non-nested regressions of h -day-ahead cumulative changes and cannot be spliced into a daily series in the same way the one-step-ahead forecast can. Subsequently, this counterfactual exercise is based on the $h = 0$ regressions alone. Given this regression, I reconstruct two counterfactual paths: one where the Odyssean shock is set to zero in the scale equation and one where it is set to zero in both equations.

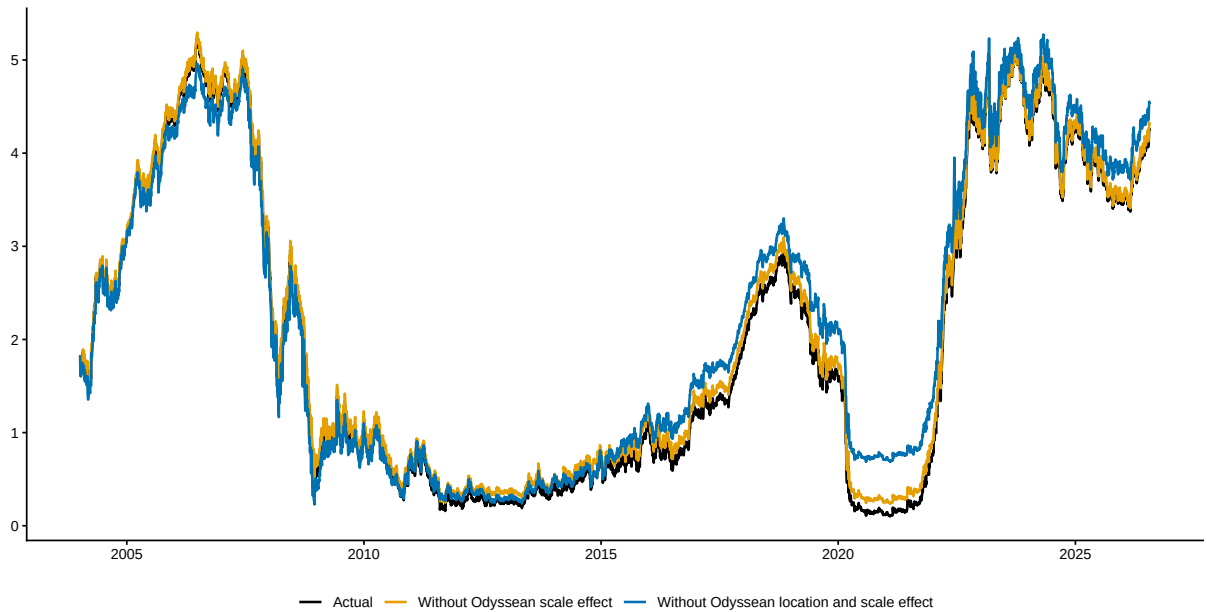


Figure 5: Actual and counterfactual 2-year nominal yield, absent the scale effect of Odyssean forward guidance and absent Odyssean forward guidance entirely.

Figure 5 reports the resulting three paths.⁵ The counterfactual switching off the Odyssean shock entirely varies considerably over time. The difference between the actual and this counterfactual series is virtually non-existent during the acute phase of the Global Financial Crisis, when explicit forward guidance was not yet a standard element of the Federal Reserve’s communication toolkit. It picks up slightly after 2015, when rates were coming off the ZLB, but the Fed was still maintaining communication that its stance is accommodative. The gap is largest over 2020-2022 when rates were again near the ZLB. Absent Odyssean guidance, the counterfactual 2-year yield runs consistently above its actual level throughout the acute phase of the pandemic and its immediate aftermath.

Consistent with the earlier findings, the scale effects are considerably smaller than that of the location. The majority of the differences between the actual and counterfactual series are driven by the location effects of the Odyssean shock. However, during some periods, particularly after rates lowered to the ZLB following the Covid pandemic, the scale channel accounts for a few basis points of difference too. These results are consistent with the Federal Reserve’s explicit, outcome-based commitments to hold the policy rate near zero having been effective at anchoring near-term rate expectations during a period of exceptional uncertainty. As minor as the scale effects are, they do appear to materially translate to a reduction in the pricing of interest rate risk, at least during the period following the pandemic.

5 Conclusions

This paper provides a comprehensive distributional perspective on how monetary policy and forward guidance shape the U.S. term structure of interest rates. By integrating high-frequency identification into a dynamic location-scale model, the analysis demonstrates that FOMC announcements do more than shift the expected path of future interest rates, they also affect the conditional volatility that investors face when pricing and hedging interest-rate risk. Standard policy rate interventions tend to heighten market uncertainty across the yield curve, whereas central bank communication—both in the form of Odyssean commitments and Delphic information signals—serves as a vital tool for compressing volatility and coordinating market expectations.

The findings highlight that the transmission mechanisms of these shocks are distinct and complex. Delphic guidance effectively reduces uncertainty across both expected short rates and term premia, and its effects spill over primarily into the volatility of inflation expectations. Conversely, Odyssean guidance coordinates medium-term expectations regarding real rates but can inadvertently increase long-term uncertainty. The results reveal that managing the volatility of expectations requires the central bank to release superior information directly in order to “calm down” the markets. The results also reveal two trade-offs. In the short run, policy rate shocks that are needed to cool the economy seem to raise uncertainty. In the medium-run, commitments to keep rates low requires giving up some certainty in the evolution of longer

⁵This exercise is reported for the Odyssean shock alone. This is done because, as Figure 1 showed earlier, the $h = 0$ impulse response coefficients are near zero for both the Target and Delphic shock and only build up gradually over the following days. Consequently, a counterfactual constructed purely from their $h = 0$ coefficients would mechanically show almost no divergence from the actual series. Conversely the coefficients are sizable at $h = 0$ for the Odyssean shock, which makes this exercise informative about this monetary policy instrument.

run yields. The results explicitly measure these in financial markets, however, implicitly this might spill over to the real economy.

Ultimately, focusing solely on the conditional mean risks missing a critical part of the monetary policy transmission mechanism. Because fixed-income markets price both risk and expectations, the systematic reductions in conditional volatility achieved through effective forward guidance can materially influence term premia, portfolio allocation, and broader financial conditions. Future research could extend this distributional framework to analyze alternative event windows, such as press conferences and meeting minutes, to further disentangle how various modes of central bank communication shape macroeconomic risks. Additionally, future research could address if the volatility channel exists beyond financial market data, and as such whether monetary policy can impact forecasters' uncertainty with respect to key macro aggregates.

6 References

- Acosta, Miguel, Andrea Ajello, Michael D Bauer, Francesca Loria, and Silvia Miranda-Agrippino. 2025. "Financial Market Effects of FOMC Communication: Evidence from a New Event-Study Database." IMFS Working Paper Series.
- Adrian, Tobias, Nina Boyarchenko, and Domenico Giannone. 2019. "Vulnerable Growth." *American Economic Review* 109 (4): 1263–89.
- Andrade, Philippe, Gaetano Gaballo, Eric Mengus, and Benoit Mojon. 2019. "Forward Guidance and Heterogeneous Beliefs." *American Economic Journal: Macroeconomics* 11 (3): 1–29.
- Angrist, Joshua D, Òscar Jordà, and Guido M Kuersteiner. 2018. "Semiparametric Estimates of Monetary Policy Effects: String Theory Revisited." *Journal of Business & Economic Statistics* 36 (3): 371–87.
- Baker, Scott R, Nicholas Bloom, and Steven J Davis. 2016. "Measuring Economic Policy Uncertainty." *The Quarterly Journal of Economics* 131 (4): 1593–1636.
- Bauer, Michael D, and Eric T Swanson. 2023. "An Alternative Explanation for the 'Fed Information Effect'." *American Economic Review* 113 (3): 664–700.
- Bernanke, Ben S, and Kenneth N Kuttner. 2005. "What Explains the Stock Market's Reaction to Federal Reserve Policy?" *The Journal of Finance* 60 (3): 1221–57.
- Bloom, Nicholas. 2009. "The Impact of Uncertainty Shocks." *Econometrica* 77 (3): 623–85.
- Bollerslev, Tim. 1986. "Generalized Autoregressive Conditional Heteroskedasticity." *Journal of Econometrics* 31 (3): 307–27.
- Buckman, Shelby R, Adam Hale Shapiro, Moritz Sudhof, Daniel J Wilson, et al. 2020. "News Sentiment in the Time of COVID-19." *FRBSF Economic Letter* 8 (1): 5–10.
- Bundick, Brent, Trenton Herriford, and A Lee Smith. 2024. "The Term Structure of Monetary Policy Uncertainty." *Journal of Economic Dynamics and Control* 160: 104803.
- Bundick, Brent, Trenton Herriford, and Andrew Smith. 2017. "Forward Guidance, Monetary Policy Uncertainty, and the Term Premium." *Monetary Policy Uncertainty, and the Term Premium (July 2017)*.
- Campbell, Jeffrey R, Charles L Evans, Jonas DM Fisher, Alejandro Justiniano, Charles W Calomiris, and Michael Woodford. 2012. "Macroeconomic Effects of Federal Reserve Forward Guidance [with Comments and Discussion]." *Brookings Papers on Economic Activity*, 1–80.
- Christensen, Jens HE, Francis X Diebold, and Glenn D Rudebusch. 2011. "The Affine Arbitrage-Free Class of Nelson–Siegel Term Structure Models." *Journal of Econometrics* 164 (1): 4–20.
- Cover, James Peery. 1992. "Asymmetric Effects of Positive and Negative Money-Supply Shocks." *The Quarterly Journal of Economics* 107 (4): 1261–82.
- Engle, Robert F. 1982. "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation." *Econometrica: Journal of the Econometric Society*, 987–1007.
- Filardo, Andrew J, and Boris Hofmann. 2014. "Forward Guidance at the Zero Lower Bound." *BIS Quarterly Review March*.
- Frangiamore, Francesco, Davide Furceri, Domenico Giannone, Faizaan Kisat, and Pietro Pizzuto. 2025. "The Effects of Fiscal Consolidations on the Debt Distribution."

- Furceri, Davide, Domenico Giannone, Faizaan Kisat, W Raphael Lam, and Hongchi Li. 2025. “Debt-at-Risk.” International Monetary Fund.
- Gurkaynak, Refet S, Brian Sack, and Eric T Swanson. 2005. “Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements.”
- Gürkaynak, Refet S, Brian Sack, and Jonathan H Wright. 2007. “The US Treasury Yield Curve: 1961 to the Present.” *Journal of Monetary Economics* 54 (8): 2291–2304.
- Hanson, Samuel G, and Jeremy C Stein. 2015. “Monetary Policy and Long-Term Real Rates.” *Journal of Financial Economics* 115 (3): 429–48.
- Hubert, Paul, and Fabien Labondance. 2018. “The Effect of ECB Forward Guidance on the Term Structure of Interest Rates.” *International Journal of Central Banking* 14 (5): 193–222.
- Husted, Lucas, John Rogers, and Bo Sun. 2020. “Monetary Policy Uncertainty.” *Journal of Monetary Economics* 115: 20–36.
- Jarocinski, Marek, and Peter Karadi. 2025. “Disentangling Monetary Policy, Central Bank Information, and Fed Response to News Shocks.” *CEPR Discuss. Pap* 19923.
- Jarociński, Marek. 2024. “Estimating the Fed’s Unconventional Policy Shocks.” *Journal of Monetary Economics* 144: 103548.
- Jarociński, Marek, and Peter Karadi. 2020. “Deconstructing Monetary Policy Surprises—the Role of Information Shocks.” *American Economic Journal: Macroeconomics* 12 (2): 1–43.
- Jordà, Òscar. 2005. “Estimation and Inference of Impulse Responses by Local Projections.” *American Economic Review* 95 (1): 161–82.
- Jurado, Kyle, Sydney C Ludvigson, and Serena Ng. 2015. “Measuring Uncertainty.” *American Economic Review* 105 (3): 1177–1216.
- Krippner, Leo. 2013. “Measuring the Stance of Monetary Policy in Zero Lower Bound Environments.” *Economics Letters* 118 (1): 135–38.
- Kuttner, Kenneth N. 2001. “Monetary Policy Surprises and Interest Rates: Evidence from the Fed Funds Futures Market.” *Journal of Monetary Economics* 47 (3): 523–44.
- Lopez-Salido, David, and Francesca Loria. 2024. “Inflation at Risk.” *Journal of Monetary Economics* 145: 103570.
- Machado, José AF, and JMC Santos Silva. 2019. “Quantiles via Moments.” *Journal of Econometrics* 213 (1): 145–73.
- Nakamura, Emi, and Jón Steinsson. 2018. “High-Frequency Identification of Monetary Non-Neutrality: The Information Effect.” *The Quarterly Journal of Economics* 133 (3): 1283–1330.
- Rigby, Robert A, and D Mikis Stasinopoulos. 2005. “Generalized Additive Models for Location, Scale and Shape.” *Journal of the Royal Statistical Society Series C: Applied Statistics* 54 (3): 507–54.
- Rogers, John H, Chiara Scotti, and Jonathan H Wright. 2018. “Unconventional Monetary Policy and International Risk Premia.” *Journal of Money, Credit and Banking* 50 (8): 1827–50.
- Swanson, Eric T. 2021. “Measuring the Effects of Federal Reserve Forward Guidance and Asset Purchases on Financial Markets.” *Journal of Monetary Economics* 118: 32–53.
- Tenreyro, Silvana, and Gregory Thwaites. 2016. “Pushing on a String: US Monetary Policy Is Less Powerful in Recessions.” *American Economic Journal: Macroeconomics* 8 (4): 43–74.

7 Appendix

7.1 Data and descriptives

Table 2 summarizes the monetary policy shocks identified using the alternative HFI approaches considered in this paper. All shock series are replicated using the USMPD database of Acosta et al. (2025), specifically the high-frequency changes observed in 30-minute windows around FOMC statement releases. For each identification scheme, I closely follow the original paper together with its accompanying replication files in selecting the relevant financial variables and implementing the estimation procedure. To facilitate comparisons across both shock series and identification methods, all shocks are scaled to have

Variable	Mean	SD	Min	Max	Skewness	Kurtosis
<i>Nominal Yields</i>						
N6M	3.217	2.468	0.05	9.702	0.285	1.997
N1Y	3.31	2.482	0.055	9.802	0.279	2.015
N2Y	3.498	2.442	0.102	9.733	0.284	2.057
N3Y	3.682	2.38	0.127	9.659	0.303	2.102
N5Y	4.018	2.271	0.222	9.541	0.338	2.18
N10Y	4.63	2.132	0.52	9.43	0.304	2.26
<i>Expected Short Rates</i>						
ESR6M	2.312	2.039	-0.224	6.569	0.451	1.673
ESR1Y	2.324	1.979	-1.227	6.582	0.398	1.698
ESR2Y	2.397	1.842	-2.482	6.479	0.315	1.821
ESR3Y	2.489	1.696	-2.832	6.336	0.292	1.978
ESR5Y	2.654	1.446	-2.443	6.085	0.36	2.258
ESR10Y	2.889	1.139	-0.625	5.754	0.578	2.645
<i>Term Premia</i>						
TP6M	0.006	0.224	-0.402	1.298	1.114	4.662
TP1Y	0.036	0.369	-0.68	2.338	0.941	4.823
TP2Y	0.1	0.571	-1.124	3.649	0.923	4.811
TP3Y	0.172	0.688	-1.428	4.364	0.873	4.739
TP5Y	0.34	0.812	-1.723	5.017	0.723	4.392
TP10Y	0.753	0.915	-1.576	5.222	0.559	3.453
<i>TIPS Yields</i>						
T2Y	0.942	1.722	-2.274	7.304	0.583	3.271
T3Y	1.191	1.601	-1.907	6.095	0.398	2.758
T5Y	1.57	1.381	-1.161	4.62	0.182	2.449
T10Y	2.077	1.101	-0.417	4.352	-0.137	2.166
<i>Breakeven Inflation</i>						
B2Y	1.783	0.757	-2.515	4.084	-1.71	7.158
B3Y	1.901	0.571	-1.109	2.912	-1.599	6.093
B5Y	2.168	0.365	0.694	3.03	-0.581	2.896
B10Y	2.487	0.523	0.887	3.876	-0.166	2.399

Table 1: Descriptive Statistics: Yield Curve Components

unit standard deviation. The identification procedures of Rogers et al. (2018) and Jarociński (2024) additionally recover an LSAP factor, which is included as a control variable in the empirical analysis. Since the focus of this paper is primarily on forward guidance, the responses to the LSAP shock are not reported.

Variable	Mean	SD	Min	Max	Skewness	Kurtosis
<i>Jarociński (2024)</i>						
Target	-0.152	1	-7.477	1.987	-3.961	26.314
Odyssean	-0.034	1	-4.32	3.462	-0.2	5.866
LSAP	-0.017	1	-13.031	2.517	-8.55	112.356
Delphic	-0.067	1	-4.077	6.815	0.701	12.498
<i>Rogers et al. (2018)</i>						
Target	-0.155	1	-6.883	2.418	-3.338	21.287
Path	-0.066	1	-3.692	4.134	-0.12	5.614
LSAP	-0.022	1	-13.307	2.358	-8.558	114.328
<i>Jarociński and Karadi (2020)</i>						
MP	-0.019	1	-6.512	2.342	-2.212	14.585
CBI	0.012	1	-3.771	4.734	-0.351	6.408

Table 2: Descriptive Statistics: Monetary Policy Shocks

For the identification strategy of Rogers et al. (2018), the information set consists of the first federal funds futures contract (MP1), the fourth Eurodollar futures contract (ED4), and the 10-year Treasury yield (ONRUN10). MP1 serves as the Target factor, ED4 orthogonalized with respect to MP1 serves as the Path factor, and the LSAP factor is ONRUN10 orthogonalized to Target and Path surprises and is set to 0 prior to October of 2008. Following Jarociński and Karadi (2020), the first principal component (PC1) of the first and fourth federal funds futures contracts (MP1 and FF4) together with the second through fourth Eurodollar futures contracts (ED2, ED3 and ED4) capture market responses to FOMC announcements. Along PC1, high-frequency changes in the S&P 500 (SP500) are used to recover the MP and CBI shocks such that MP raises interest rates and depresses stock returns, while CBI raises rates and stock returns. The shock series are recovered using the median-rotation algorithm. Finally, the identification scheme of Jarociński (2024) uses the first federal funds futures contract (MP1), the 2-year and 10-year Treasury yields (ONRUN2 and ONRUN10), and high-frequency changes in the S&P 500 returns (SP500). These variables are used to recover four structural shocks corresponding to Target rate surprises, Odyssean forward guidance, Delphic information, and an LSAP factor.

Given that the monetary policy shocks are identified using state-of-the-art HFI methods, additional control variables are not required for the consistent estimation of the average effects of monetary policy and forward guidance on the yield curve.⁶ Nevertheless, I include a broad set of macro-financial controls to account for co-movements that may contribute to yield dynamics. In the location-scale models, these controls help isolate the residual variation used to construct the volatility proxy. In practice, however, their inclusion has only a negligible effect on both the estimated location and scale responses.

The choice of control variables closely follows the specification of Hubert and Labondance (2018), with some extensions and adapted to the U.S. economy and financial markets. These variables capture mon-

⁶For example, the long-run daily local projections in Jarociński (2024) are estimated without additional controls.

Variable	Mean	SD	Min	Max	Skewness	Kurtosis
<i>Controls</i>						
CISS	0.092	0.139	0	0.896	2.875	12.909
WTI	0	0.027	-0.424	0.426	-0.245	34.242
SP500	0	0.012	-0.128	0.11	-0.382	13.753
USD	4.526	0.114	4.267	4.789	-0.019	2.297
VIX	19.901	8.057	9.14	82.69	2.169	11.301
EPU	116.914	98.118	3.32	1026.38	2.479	11.756
NSI	0.023	0.188	-0.666	0.454	-0.57	3.21
SSR	2.008	2.859	-4.169	6.872	-0.234	1.911

Table 3: Descriptive Statistics: Control Variables

etary policy stance, financial market conditions, uncertainty, and economic sentiment. Specifically, the Shadow Short Rate (SSR) of Krippner (2013) captures the stance of monetary policy, particularly during periods in which the federal funds rate is constrained by the effective lower bound. The ECB’s Composite Indicator of Systemic Stress (CISS) provides a broad measure of systemic stress in the financial system. Changes in the S&P 500 index and WTI crude oil prices proxy for stock market sentiment and inflation shocks. The News Sentiment Index of Buckman et al. (2020) provides a broad, news based economic sentiment measure. The VIX index and the Economic Policy Uncertainty (EPU) of Baker et al. (2016) provide uncertainty measures. Lastly, the NYSE US Dollar Index (USD) is included to account for the international valuation of the US Dollar.

7.2 Robustness and sensitivity checks

To assess the robustness of the results, I consider three modifications to the baseline specification: first, I estimate the model without the ARCH term in the scale equation;⁷ second, I drop the control variables that capture macro-financial fluctuations; third, I replace the baseline shocks identified using the methodology of Jarociński (2024) with alternative high-frequency identification schemes from the literature. Specifically, the path factor of Rogers et al. (2018) serves as an alternative measure of Odyssean forward guidance, while the central bank information shock of Jarociński and Karadi (2020) provides an alternative proxy for the Delphic component. To ensure comparability across specifications, these alternative shock series are also reconstructed using the USMPD database, thereby preserving the same sample period throughout the analysis.

⁷In principle, this robustness exercise could also consider higher-order ARCH specifications. However, including additional lags yields impulse responses that are virtually identical to those obtained with a single lag. These specifications are therefore omitted, as they provide little additional insight into the sensitivity of the results to volatility clustering.

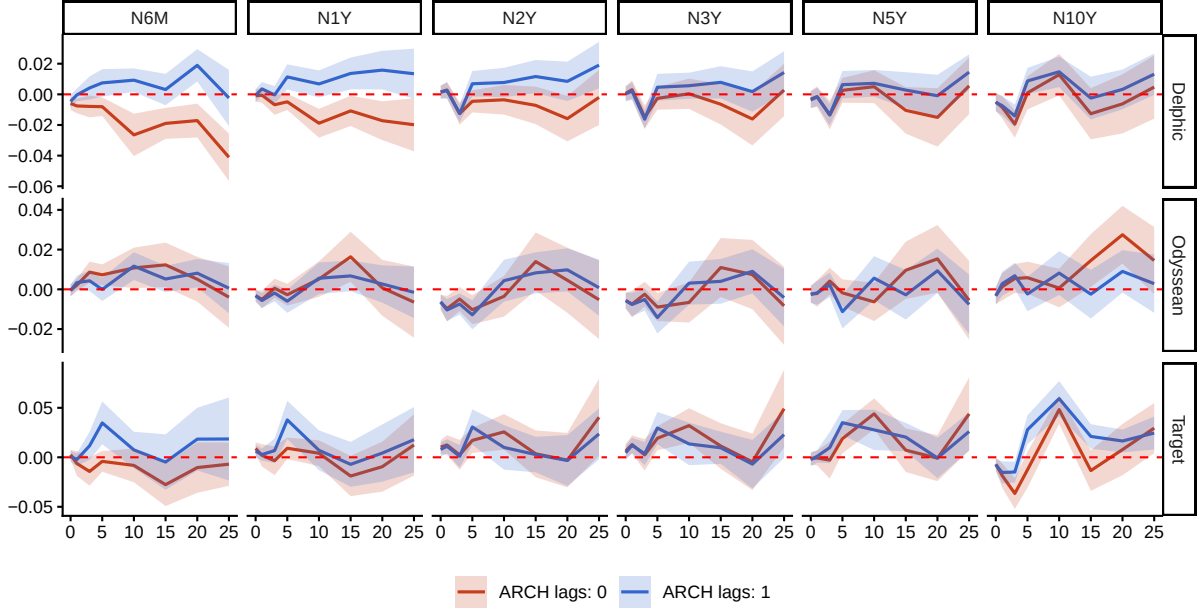


Figure 6: Sensitivity of the scale estimates to controlling for volatility clustering.

Figures 6 and 7 compare the estimated scale impulse responses obtained with and without the ARCH term in the scale equation and with dropping the macro-financial controls respectively. Overall, the results are highly robust to accounting for volatility clustering. In particular, the responses at longer maturities and shorter impulse response horizons remain largely unchanged, while the contemporaneous ($h = 0$) responses are identical under both specifications. The main difference arises for the Delphic (central bank information) shock, particularly at the short end of the yield curve, where omitting the ARCH term leads to somewhat larger estimated reductions in conditional volatility. Regarding the use of control variables, the results are robust to specifications with or without using them. This further reinforces the ARCH specification, as it alone absorbs most of the variation of absolute residuals that would otherwise be captured by other variables.

The sensitivity of the results to accounting for volatility clustering is economically intuitive. As reported in Table 1, the unconditional standard deviation is substantially higher at the short end of the yield curve than at longer maturities. While long-term yields are driven to a greater extent by macroeconomic fundamentals, short-term yields are more susceptible to short-run fluctuations in expectations and financial market conditions. Consequently, failing to account for volatility clustering causes part of the persistence in short-term yield volatility to be attributed to the Delphic (information) shock. At longer maturities, this bias is smaller and the qualitative conclusions remain unchanged. At the short end of the yield curve, however, omitting the ARCH term leads to a noticeable overstatement of the estimated volatility-compressing effect.

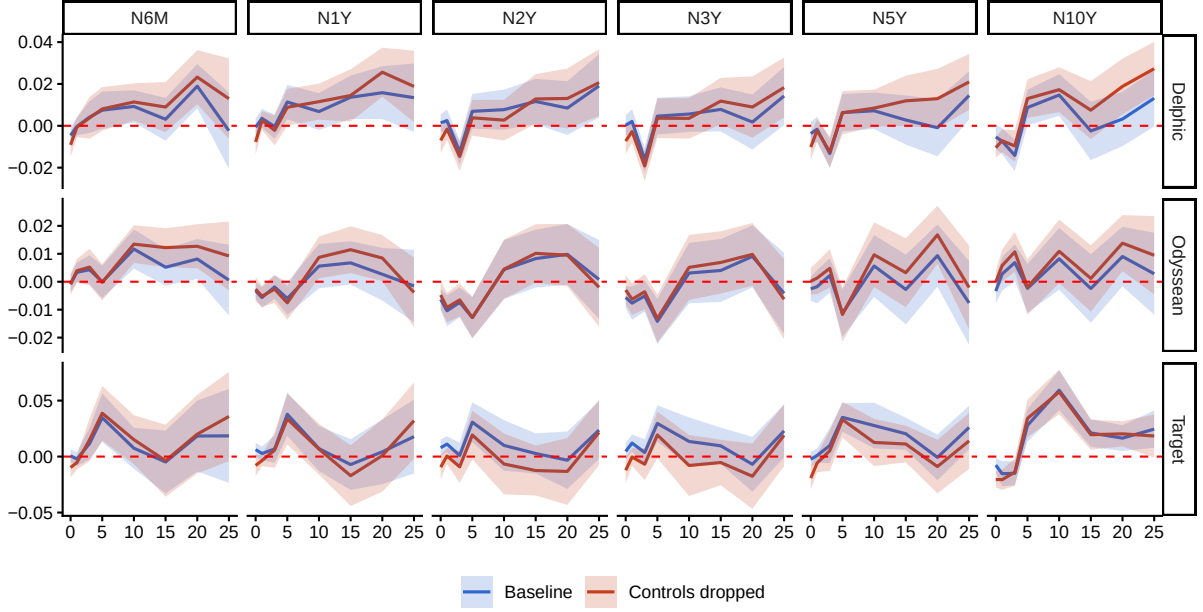


Figure 7: Sensitivity of the scale estimates to dropping control variables.

Figures 8 and 9 show the dynamic location and scale responses obtained using the identification schemes of Rogers et al. (2018) and Jarociński and Karadi (2020) respectively. In the former, both the location and scale responses of the Target and Path factors are virtually identical to those of the Target and Odyssean shocks reported earlier in Figure 1. In the latter, the location and scale responses of the term structure to a CBI shock match closely to those obtained with the Delphic shock. On the other hand, responses to the MP shock appear to reflect signals from Target, Odyssean (and possibly LSAP) surprises. Consequently, the results appear to be robust to the choice of orthogonalization and high-frequency series used to construct the surprises.

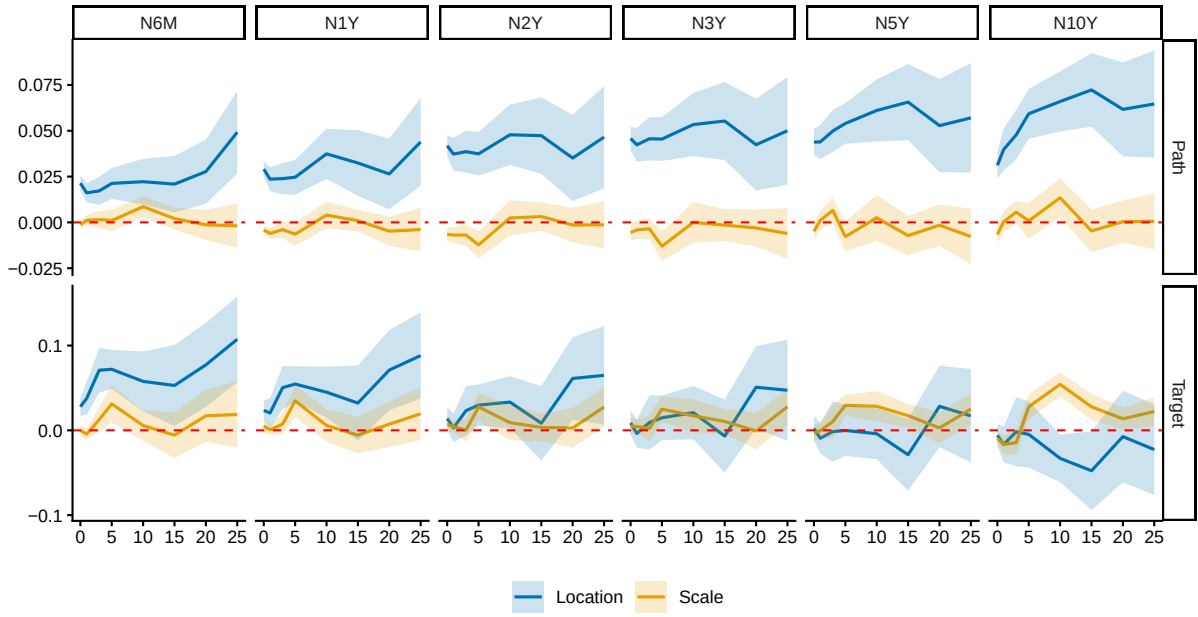


Figure 8: Dynamic responses of the nominal term structure - using the shock identification of Rogers et al. (2018)

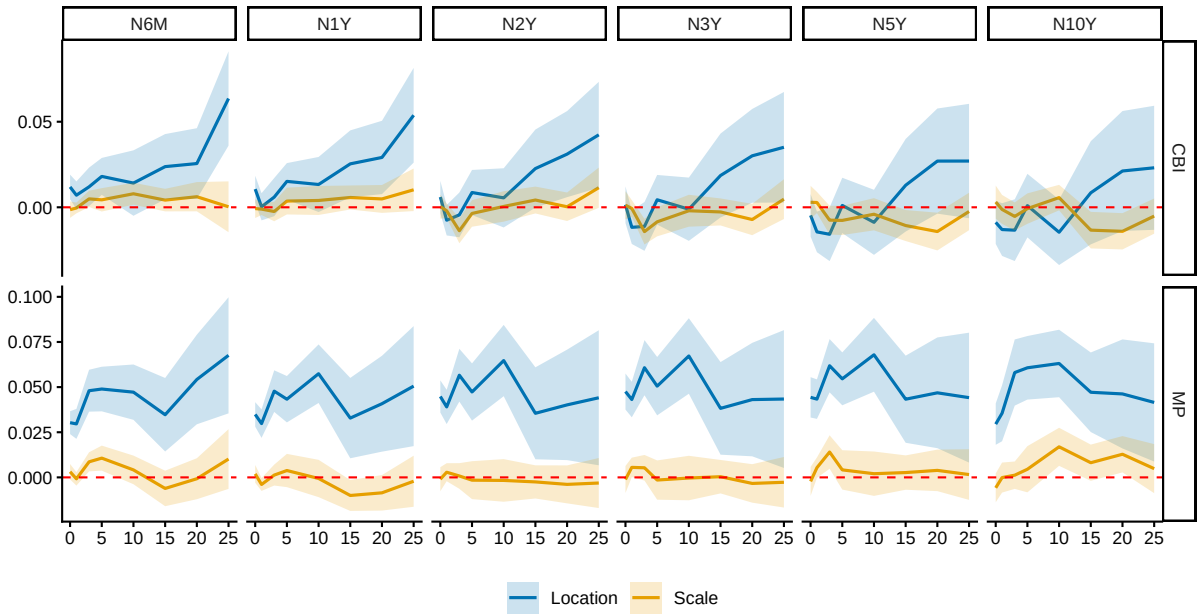


Figure 9: Dynamic responses of the nominal term structure - using the shock identification of Jarociński and Karadi (2020)